Computer Vision

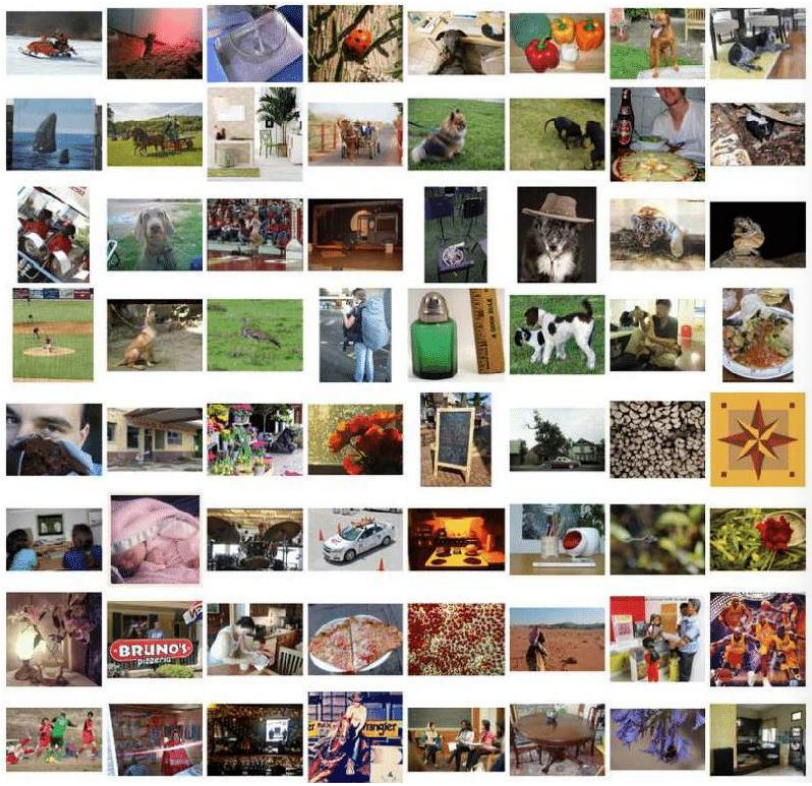
DPS920

Session 22

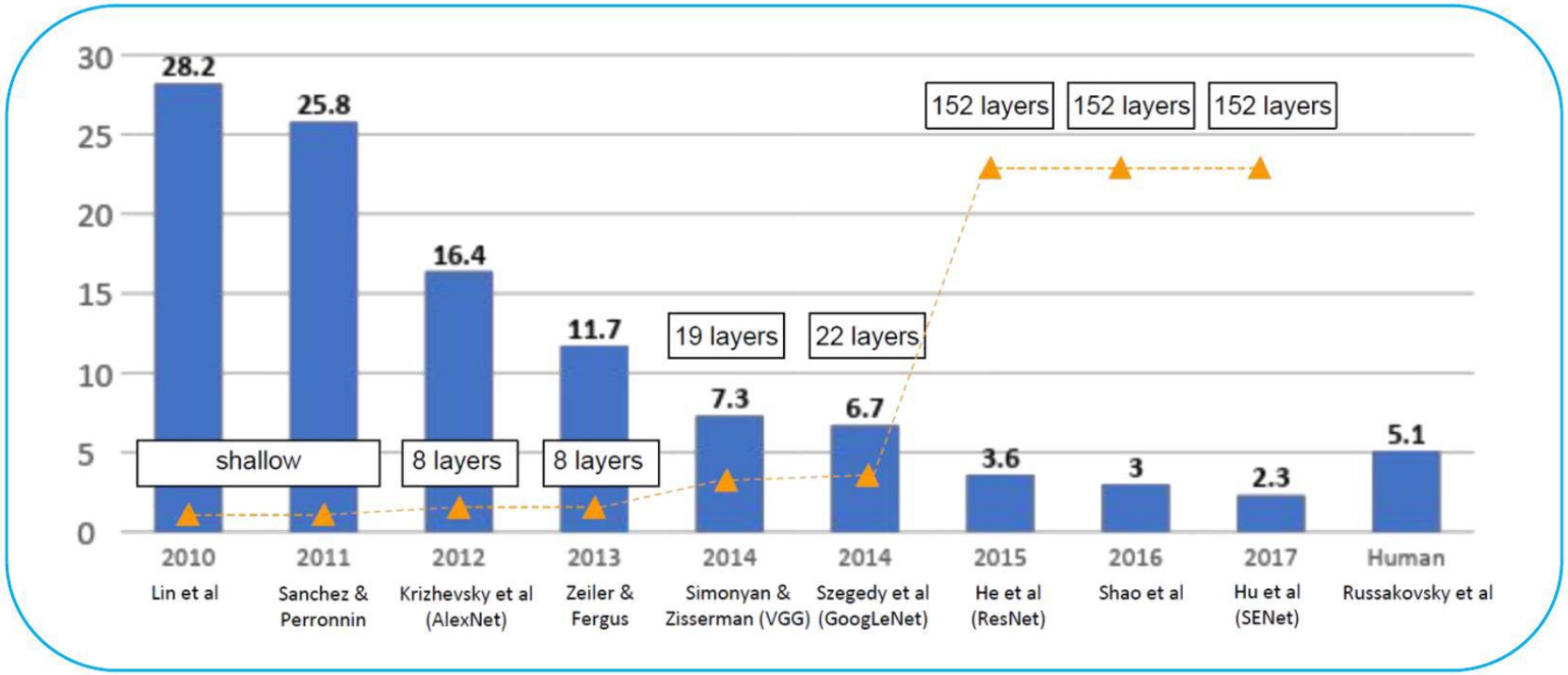
ILSVRC Challenge (2010-2017)

- ImageNet dataset (14 M)
- 1.2M images with 1000 classes

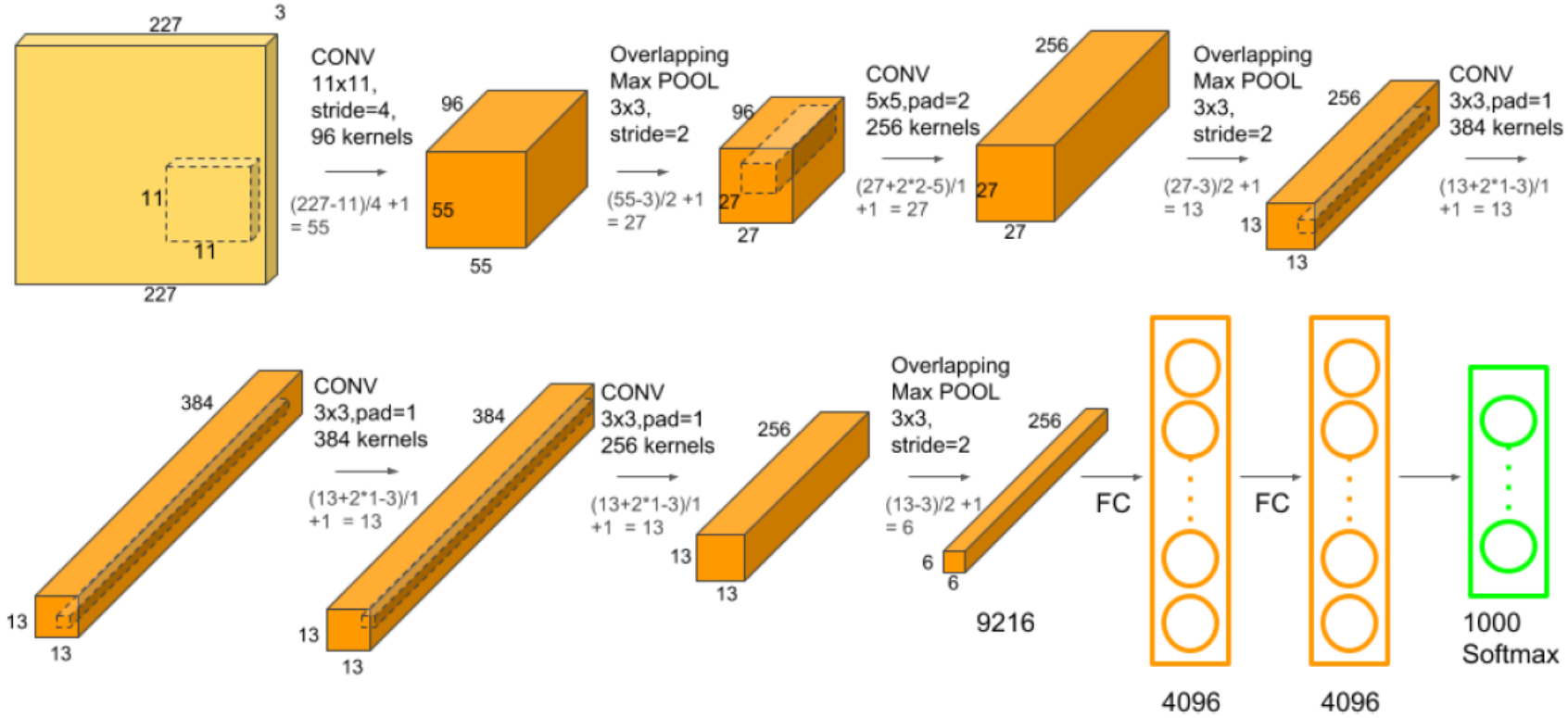
- Classification
- Single object detection
- Object detection



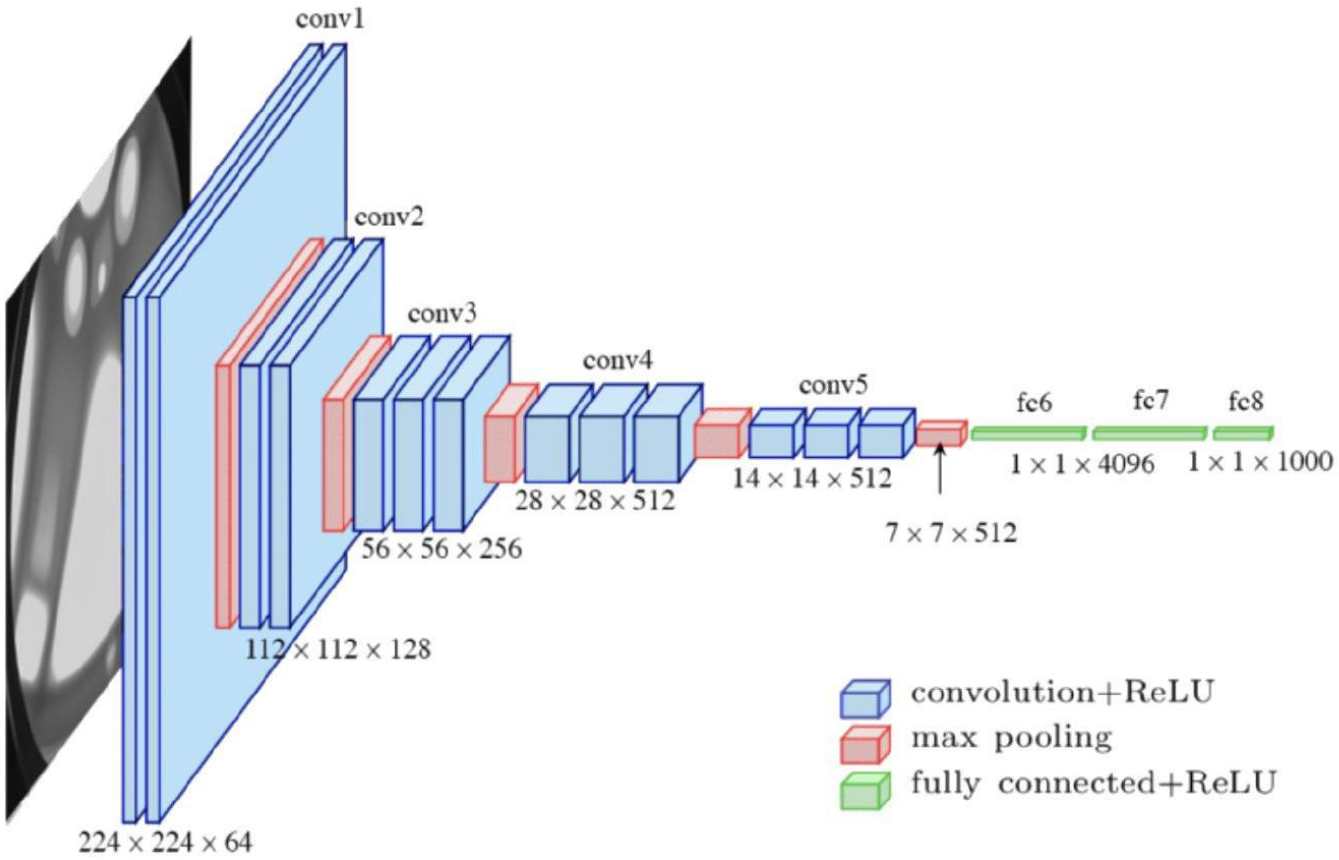
Results


30
25
20
15
10
5
0
28.2
25.8
16.4
11.7
7.3
6.7
3.6
3
2.3
5.1
shallow
8 layers
8 layers
19 layers
22 layers
152 layers
152 layers
152 layers
2010
Lin et al
2011
Sanchez & Perronnin
2012
Krizhevsky et al (AlexNet)
2013
Zeiler & Fergus
2014
Simonyan & Zisserman (VGG)
2014
Szegedy et al (GoogLeNet)
2015
He et al (ResNet)
2016
Shao et al
2017
Hu et al (SENet)
Human
Russakovsky et al

AlexNet


227
3
11
11
227
CONV
11x11,
stride=4,
96 kernels
(227-11)/4 +1
= 55
96
55
55
Overlapping
Max POOL
3x3,
stride=2
(55-3)/2 +1
= 27
96
27
27
CONV
5x5,pad=2
256 kernels
(27+2*2-5)/1
+1 = 27
256
27
27
Overlapping
Max POOL
3x3,
stride=2
(27-3)/2 +1
= 13
256
13
13
CONV
3x3,pad=1
384 kernels
(13+2*1-3)/1
+1 = 13
384
13
13
CONV
3x3,pad=1
384 kernels
(13+2*1-3)/1
+1 = 13
384
13
13
CONV
3x3,pad=1
256 kernels
(13+2*1-3)/1
+1 = 13
256
13
13
Overlapping
Max POOL
3x3,
stride=2
(13-3)/2 +1
= 6
256
6
6
9216
FC
4096
FC
4096
1000
Softmax

VGG


conv1
conv2
conv3
conv4
conv5
fc6
fc7
fc8
224 × 224 × 64
112 × 112 × 128
56 × 56 × 256
28 × 28 × 512
14 × 14 × 512
7 × 7 × 512
1 × 1 × 4096
1 × 1 × 1000
convolution+ReLU
max pooling
fully connected+ReLU

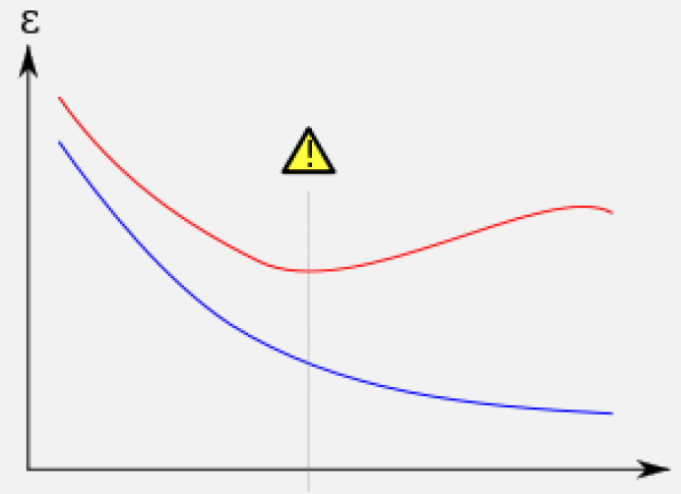
Learning Rate

A hyperparameter that controls how much the model updates its weights in response to the loss gradient.

Too high → Model diverges
Too low → Slow convergence
Just right → Fast & stable training

```
from keras.optimizers import SGD
...
opt=SGD(lr=..., decay=...)
net.compile(..., optimizer=opt)
```

Overfitting



- Model works good on train data but not on test data
- It memorizes and not generalize

Batch Normalization

- We normalized the input data.
- Why not normalizing values after each layer?

- A technique to normalize activations in a neural network, layer by layer, during training.

- Stabilizes learning
- Speeds up convergence
- Reduces internal covariate shift
- Allows higher learning rates

Data Augmentation

- A technique to increase the size and diversity of training data by applying random transformations.
- More robust models with better performance on unseen data.

Rotation

Width Shift

Brightness

Shear

Zoom

ImageDataGenerator

A tool to apply data augmentation

```
aug = ImageDataGenerator(rotation_range=30,
                         width_shift_range=0.1,
                         height_shift_range=0.1,
                         shear_range=0.2,
                         zoom_range=0.2,
                         horizontal_flip=True,
                         fill_mode="nearest")
```